

Exercise Set 1

Exercise 1

- a) I chose MacOS as my operating system. As it is the one I am using on my personal computer.
- b) Running processes in MacOS can be viewed using the “Activity Monitor” application.
- c) I chose “Code” or Visual Studio Code as my process. Relevant information includes:
 - Name: Code
 - User: dmitriigorovoi
 - Process ID (PID): 54820
 - CPU usage: 4,1%
 - CPU time: 19.21
 - Memory usage: 125,8 MB
 - Number of threads: 46

Exercise 2

- a) When process is created this happens:
 - Assign an identity: create a new process ID (PID).
 - Create and initialize the process control block (PCB).
 - Set up the execution context: initialize the program counter, registers, stack pointer, and initial CPU context so it can run.
 - Provide resources and address space: create the process address space , set up memory mappings, and apply limits/permissions.
 - Inherit or attach OS objects: copy or share things like open files, current directory, environment, credentials, and IPC handles depending on the creation method.
 - Make it schedulable: set state to Ready and place it into the appropriate scheduler queue(s) so it can be dispatched.
- b)
 - Normal completion: The program finishes its work and calls an exit routine

- Error or exception: The process hits a runtime error the OS cannot safely recover from, like an illegal instruction, segmentation fault, divide-by-zero, or failed protection check.
- Killed by another entity: A user, administrator, parent process, or the OS requests termination